Measuring tropospheric gravity waves over stratocumulus cloud decks

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Key Points:

- A 2D wavelet method distinguishes intrinsic propagation from trade-wind advection in stratocumulus clouds in infrared geostationary imagery.
- Identified wave events are tracked as elliptical packets using their persistence over many hours.
- A cloud-clearing low-level divergence wavefront over the SE Pacific, generated by a midlatitude jet trough, is also seen in reanalysis.

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Abstract

Internal gravity waves modulate marine stratocumulus clouds, but subtly, comprising only a small part of the space-time variance in geostationary satellite animations. To isolate them, we apply a two-dimensional continuous wavelet transform that seeks out packets of parallel stripes moving at intrinsic phase velocity sufficiently different from the background wind. This methodology enables both comprehensive regional scanning and high-resolution tracking of individual wave packets. This paper presents a strong event detected in a 3-month dataset over the southeast Pacific, traceable back to an upper-tropospheric trough and jet streak in ERA5 reanalysis data. Using the divergence of wind fields from low-cloud tracking, dynamical and optical wave amplitudes can be simultaneously estimated from satellite imagery alone. This capability may shed light on other subtle phenomena, such as vigorous wave sources hidden in cloudy storms and complex geography, and might even become an assimilation-relevant input to weather models.

Plain Language Summary

Gravity waves are invisible undulations within the atmosphere, generated by vigorous weather and other complex processes. Over the oceans, these waves travel above large sheets of low-level clouds, slightly raising and lowering the cloud tops, which changes the cloud brightness seen by weather satellites. However, measuring these waves is difficult because the clouds are simultaneously churning in complex ways and being blown by the wind. Our new mathematical methods discover and track these waves over time. A strong cloud-clearing wave over the Southeast Pacific is shown, emitted by a midlatitude storm. These new observational tools can challenge and improve atmospheric models by spotlighting subtle sources of imbalance in the atmosphere.

1 Introduction

As the simplest linear solutions to the governing fluid equations, internal gravity waves are ubiquitous in the stratified atmosphere. Their dispersion relation permits a wide range of frequencies and wavelengths, meaning the actual population of waves is primarily dictated by their diverse sources and propagation pathways. The primary sources include flow over orography, moist convection, and dynamical imbalances associated with jet streams and fronts (Plougonven & Zhang, 2014). Because these sources involve highly complex, non-linear, and transient processes, direct observations of the smoothly prop-

agating wave field in the far-field environment may serve as a usefully integrated diagnostic of complicated dynamics deep inside vigorous overcast weather situations. Quantifying these waves could allow satellite imagery to help distinguish which members are better or worse in ensemble data assimilation.

Direct observations hold apex importance in such a theoretically under-determined problem, yet they remain exceptionally challenging. Clear-air stratification acts cleanly as a restoring force to vertical motions, but those are almost invisible to standard optical sensors (although the vapor-channel differencing methods of Poujol and Bony 2025 are a recent breakthrough). Vertically pointing clear-air lidars occasionally capture wavy vertical displacement fields (airmass stratigraphy), but these local datasets are sparse. Spaceborne Doppler lidars can access the entire globe and reveal velocity fields, including their wavy components, wherever clouds do not block the signal (Ratynski et al., 2023, 2025). However, they provide only disconnected snapshots in space and time. Dense barograph networks demonstrate that high-frequency waves are present from the stratosphere down to the surface (de Groot-Hedlin et al., 2017), and offer excellent temporal resolution, but they are extremely sparse geographically (L. R. Allen et al., 2025).

To overcome these observational gaps, we turn to the extensive archives of satellite imagery over the vast marine stratocumulus (Sc) cloud decks capping the Planetary Boundary Layer (PBL) in the subtropics. Because these clouds cover approximately one-fifth of the Earth's surface, their delicate equilibrium is profoundly important in the global radiative budget (Wood, 2012). These extensive fields also serve as a natural visualization surface for gravity waves propagating in the stratified clear air above. The fundamental link between mesoscale gravity waves and upper-level jet streaks has been recognized for decades (Uccellini & Koch, 1987). As demonstrated by G. Allen et al. (2012), internal gravity waves of up to 400 m amplitude, generated by a disturbed subtropical jet, can propagate thousands of kilometers across the Southeast Pacific stratocumulus deck, imprinting themselves in cellular cloud textures and albedo, causing irreversible cloud clearing.

However, that work was based on manual tracking of one very large amplitude event in blurry data. We would like to objectively extract the characteristics of many such waves, and others discernible to the eye (radiating also from other sources such as continents and tropical convection and cyclones), to build up robust regional climatologies. With

enough cases, and sufficiently refined methods, we hope to enable quantitative deductions about hard to observe wave sources, the propagation medium, and downstream impacts on clouds and other weather elements.

To unlock the full potential of geostationary satellite data, we introduce an automated, objective methodology based on two-dimensional Continuous Wavelet Transforms (2D CWT). Wavelet transforms have proven effective in objectively quantifying trapped and stationary lee waves in high-resolution models, for instance (Houtman et al., 2025). Here we adapt a similar mathematical framework, applied more agnostically to survey a full range of possible wavelengths and directions. Our mother wavelet function allows wave packets to have as few as 2-3 parallel crests and troughs in a packet. Screening for a sufficiently large intrinsic phase velocity $\vec{c}_{int}$ relative to a background advecting wind $\vec{V}_{adv}$ defines a wave detection. Details are in the Methods supplement, and summarized below.

The remainder of this study is organized as follows: Sect.2 details the physical basis of the wave signatures we seek, and the two-step wavelet methodology. Sect.3 demonstrates the algorithm's capability first as a large-scale synoptic scanner, and subsequently as a tracking tool to capture the crests and troughs of a singular wave packet. Sect.4 shows a case study of one prominent wave in its broader synoptic environment. Sect.5 offers a discussion of future prospects for this capability.

2 Physical Basis and Methodology

The phenomenon we seek to detect and some key aspects of our methodology are illustrated in Figure 1.

The physical basis for detection relies on cloud "brightness" (visual or infrared) modulation, as depicted in Fig. 1a. Cloud thickening and deepening (gray in the schematic, lighter areas in imagery) occur in response to the wave's vertical displacement of the marine boundary layer (BL) inversion, depicted at 1 km. A vertical wavelength of several km is implied, so that the stratified medium is the whole stable lower troposphere, not just the thin inversion. Intrinsic propagation speeds of order 10 m s^{-1} confirm that implication, via the phase speed of linear hydrostatic waves: one vertical wavelength per buoyancy period (about 10 minutes for moist adiabatic stratification, which the low-latitude troposphere approximates).

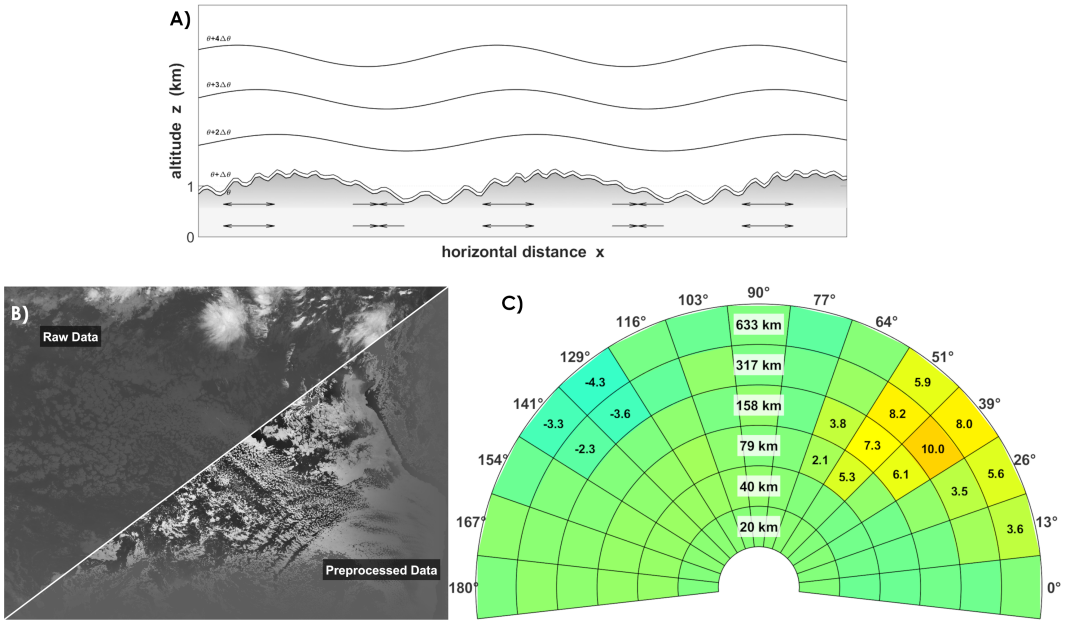


Figure 1. Overview of the wave detection methodology. (a) Schematic physical basis: a deep gravity wave, indicated by potential temperature (ϑ) surfaces, modulates the cloud-topped boundary layer depth. In crests, cloud physical and optical depth (gray shading) are greatest, while smaller-scale features are scrunched proportionately. (b) Image pre-processing: raw infrared imagery (above diagonal slash) is masked for high clouds, then spatial high-pass filtered, contrast-enhanced, and vignetted (below diagonal) to reveal mesoscale variability and avoid edge artifacts. (c) Intrinsic propagation: 2D wavelet outputs are visualized as a "Wave Rose", with wavelength and compass direction as its polar coordinates. Colors and numbers show intrinsic phase speed (that is, relative to the background flow). Orange cell with 10.0 m/s refers to a northeastward wave of 300 km wavelength on Oct. 12, 2023, 1400-2030 UTC, a speed verified in Fig. 4c.

The raising of a thin, nearly saturated layer at the BL top thickens or forms cloud, maximizing optical depth detectable in either visible or inverted-infrared imagery at wave crests. Conversely, the lowering of the BL top thins or evaporates cloud. The waves we often see visually in data animations, and target here, have horizontal wavelengths (λ) ranging from 30 to 300 km, and intrinsic phase speeds of order 10 m s^{-1}. Small-scale cellular cloud features, typically of about 30 km size (Monnier et al., 2026) have been proportionately narrowed in the crests by the same time-integrated horizontal convergence that lifted the inversion, an effect somewhat exaggerated for clarity in Fig. 1.

3 Wavelet Method for Surveying and Tracking Waves

3.1 Summary of method

To detect these subtle mesoscale modulations continuously across day and night, we rely on half-hourly window-band infrared (IR) imagery. More details of the method are in the Supporting Information (S1 and S2), but necessary key concepts are explained here.

Figure 1b illustrates the 3 preprocessing steps. First, masking nullifies high clouds (removing pixels with $T_b < 275K$ at any time in a sequence of frames being analyzed). Second, a spatial high-pass filter subtracts from each image its smoothed version using a 200-km radius Gaussian convolution. Third, the magnitude of the resulting anomaly field is square-rooted to enhance low-amplitude contrasts. This conditioning reveals the underlying mesoscale and cellular cloud patterns with contrast and clarity suited for wavelet analysis. To minimize edge artifacts in the Fourier transform step, a windowing vignette is applied near the edges as seen in Fig. 1b.

To each of these preprocessed images, we apply a 2D Continuous Wavelet Transform (CWT), using a directional 'Cauchy' mother wavelet over a logarithmic set of 6 spatial scales (roughly $16 \leq \lambda \leq 512$ km wavelengths) and 15 discrete angles (0 to 180°, as 180 to 360° are redundant). Its ouput has dimensions (NX, NY, NSCALES=5, NANGLES=15), a mixed spatial-spectral array. To speed up the spectral computation, the images are first conservatively re-gridded with a 'shrinkfactor' of 2, halving NX and NY.

Brightness spatial variance within a single frame cannot distinguish propagating waves from stationary or advecting banded cloudiness (as seen in the southeast corner of Fig. 1b for instance). To better distinguish waves, our algorithm computes the complex cross-spectrum between pairs of consecutive images. The argument of this complex product yields the phase shift ($\Delta\phi$), representing spatial translation in the cross-crests direction over the half-hourly time step Δt.

This total cross-crest temporal evolution has two sources: advection and wave propagation. To isolate the background advective flow ($\mathbf{V}_{adv}$), known to be a smooth large-scale pattern in the trade winds, we accumulate the cross-spectral density over a 6-hour temporal window (11 half-hourly pairs in a 12-image sequence), and further average spatially over each coarse 5°×5° Region of Interest (ROI). Specifically, we establish a sta-

ble large-scale advection vector $\mathbf{V}_{adv}(x, y)$ by minimizing the squared error against the theoretical advective phase shift model:

$$\Delta\phi_{adv} = \frac{2\pi}{\lambda}(\mathbf{V}_{adv} \cdot \hat{\mathbf{k}})\Delta t \tag{1}$$

where $\hat{\mathbf{k}}$ is the unit vector in the propagation direction.

With $\mathbf{V}_{adv}$ assumed constant over the 6 hours, operating frame pair by frame pair, we subtract its expected advective phase shift from the observed cross-wavelet phase shift ($\Delta\phi_{obs}$) to determine the *intrinsic phase shift* ($\Delta\phi_{int} = \Delta\phi_{obs} - \Delta\phi_{adv}$). This residual directly yields the intrinsic phase speed:

$$c_{int} = \frac{\lambda}{2\pi\Delta t}\Delta\phi_{int} \tag{2}$$

The resulting $c_{int}(S, \theta)$ for our 6 scales and 15 angles, for a chosen spatial region with a strong wave (as illustrated further below) is consolidated into the "Wave Rose" diagram of Fig. 1c. In this polar representation, the radial coordinate represents the logarithmic wavelength λ, the angular coordinate denotes propagation direction, and the color intensity corresponds to the intrinsic phase speed c_{int} (positive or negative). In Fig. 1c, a coherent gravity wave packet is seen (red colors) as a distinct, highly localized "petal" of intrinsic speed near 10 m/s toward the ENE with wavelength near 317 km. The raw intrinsic phase shifts are algorithmically un-wrapped to correct for Nyquist aliasing at small scales, and empirically dampened at the largest scales to mitigate amplified sampling noise. The complete mathematical framework underlying the complete retrieval process, is detailed in the Supporting Information (S1 and S2).

3.2 Regional Scanning or Surveying

The algorithm above has been deployed systematically over every 5x5° square in 6-hour time blocks for 3 months to create a continuous synoptic observatory reporting four times per day, affordable on 10 hours of laptop computer time. This regional scanning capability is illustrated in Figure 2.

In most squares of Figure 2, the wave roses are quiescent (greenish), indicating that brightness feature motion is well explained by $\vec{V}_{adv}$ advection. However, several distinct kinematic signatures stand above that background. In the southeast quadrant, as shown in Fig. 1c, a highly coherent, northeastward-propagating signal lights up in red ($c_{int} >$

0). The processed satellite video of this region and time period is available in the Supporting Information (S4), where this wave is clear and distinctive.

Blue signatures ($c_{int} < 0$) indicate some North-to-South intrinsic motions in the north and east parts of the image, suggestive of deep convective sources in the ITCZ and land effects from the westward bulge of South America at the right edge of the image (Supplemental figures have base maps) . However, the northern area is contaminated with south-moving high-altitude ITCZ outflow cirrus, eluding our stationary high-cloud mask. More work will be needed to tease these effects apart robustly, along with the nature of c_{int} 'dilution' in the averaging of weak signals plus noise in the complex-wavelet processing chain. For these reasons, we focus on the strong signal (red) below.

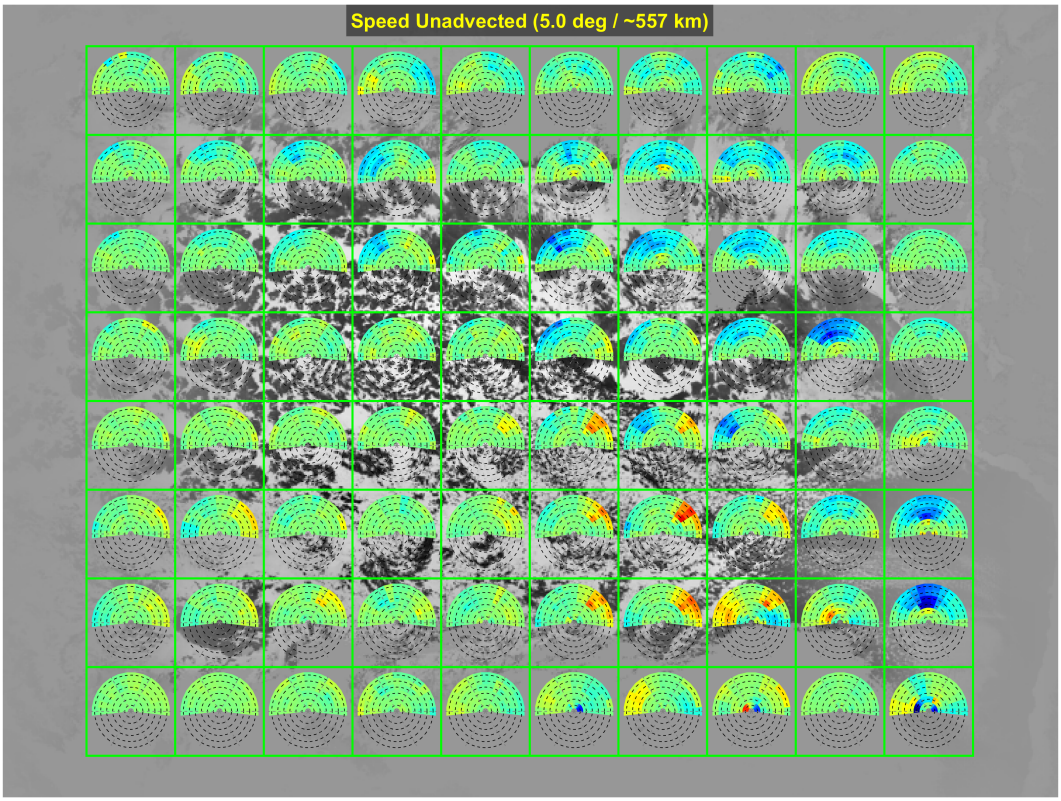


Figure 2. Large-scale survey of c_{int} wave roses over the Southeast Pacific (30.1°S to 5.1°N, 122.6°W to 77.4°W) on October 12, 2023, for the 7-hour period 14-21 UTC. Each 5x5 grid cell displays a box-averaged wave rose of intrinsic propagation speed, a difference from the background trade winds. The seventh from left, third from bottom corresponds to Fig. 1c. Plane-wave packets should show up as narrow angular beams with well-defined scale (localized 'petals' in the rose). The maximal red and blue colors correspond to 15 m/s and -15 m/s, respectively.

Spectral roses are computed by pooling complex wavelet coefficients over a specific ROI size (here $5° \times 5°$). This spatial averaging is a compromise: using a smaller ROI size increases spatial specificity but raises the noise floor. Larger ROIs provide more stable spectral estimates, but at the cost of estimation error of V_{adv} (assumed constant over the ROI), and loss of spatial localization of the wave packets we wish to isolate.

3.3 High-Resolution Tracking

Once a packet has been officially detected in this way (for instance, as a properly coherent rose petal of $|c_{int}| > 5m/s$), more detailed analysis can refine its description and localization. To home in on the true wave signal, we execute a two-phase segmentation and characterization protocol.

Intrinsic speed maps can be obtained at every pixel on every image pair within a 6-hour (12-frame) analysis block, after interpolating $\mathbf{V}_{adv}(x, y)$ from its 5-degree, 6-hour-averaged grid. The waves we seek should exhibit temporal persistence, so a Boolean array of threshold exceedance ($|c_{int}| > 5.0$ m s^{-1}) is summed over the 11 image pairs in the analysis time window. A persistent wave zone is delineated as an ellipsoidal fit to regions where that array exceeds 8 (see Supplement S2, Fig. S2). With the spatial extent defined, a focused average of the complex wavelet cross-spectrum aggregates all pixels within the ellipse and across all time steps into spectral roses of the packet, both of brighness variance density (power) and c_{int} (Fig. S3). Masking for $|c_{int}| > 5m/s$ in that spectral rose space and performing an inverse transform, that wave feature's amplitude in space-time is reconstructed in the units of (preprocessed) image brightness. For the strong wave case featured in this paper (Fig. 1c), the real part of a reconstructed wave amplitude is contoured on the images in Fig. 3

The reconstruction in Figure 3 visualizes individual crests (red) and troughs (blue) as phase lines gliding across the underlying stratocumulus texture over a 9-hour period, stitched together from two overlapping 5-hour (ten-frame) analysis blocks of half-hourly data.

Because several petals of the ellipsoid's power-rose are involved, interference makes the tracking of these contours non-steady and dependent on the spectral feature defini-

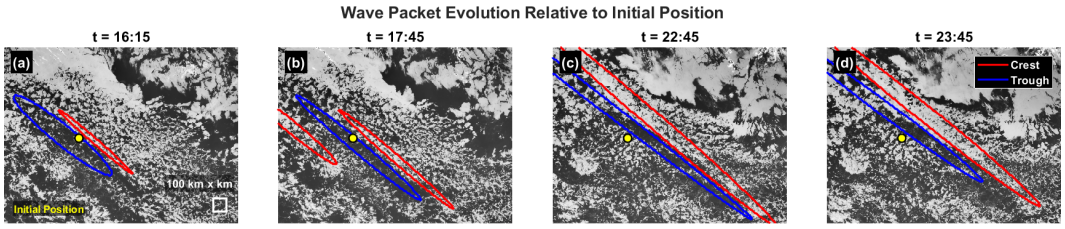


Figure 3. Raw IR imagery with contours of the tracked wavefield reconstruction for the 12 October 2023 event at the indicated hours. By applying an inverse wavelet transform based on a localized packet's spectral signature, signed brightness amplitude is the real part (contours), showing wave crests in red and troughs in blue. Whiter areas are clouds in these contrast-enhanced and sign-flipped IR images. Displayed region spans 20°S to 8°S and 104°W to 88°W.

tion. Still, these methods provide a quantitative and objective extension to the manual isochrones in the pathbreaking work of G. Allen et al. (2012). In their tracking of a similar Southeast Pacific wave event, they documented wave trains with "phase speed 15.3 m/s, period ∼1 h and horizontal wavelength 55 km". In contrast, our 12 October event is significantly longer ($\lambda \sim$ 300–360 km) and slower ($c_{int} \sim 10 m/s$).

In the hydrostatic limit, internal wave speed is set by vertical wavelength and stratification (about 1 wavelength per 10 minutes in the troposphere). For thermally forced waves, the heating profile often sets the speed, while heating event duration sets horizontal scales (Mapes, 1998). For momentum-equation forcing at a localized altitude, horizontal wavelength might be set by horizontal structure, while relative speeds in an airmass collision might set c, or a vorticity (rotation rate) might set the frequency. Waves radiated from vortices in water seem to be of this character. In a stratified flow, vertical wavelength could be determined by the dispersion relation from the source's c or wavelength, plus the frequency from a forcing timescale; or it might be set by the vertical depth of forcing, as in the thermally forced case. These interpretations are quite subtle, beyond the present scope; an extensive yet still indeterminate literature review of the considerations is in Plougonven & Zhang, 2014.

Whatever the details, the synoptic -scale length of the wavefronts clearly parallels a synoptic upper-level trough, as shown next. The role of advective nonlinearities in the momentum equation vs. possible latent heating in frontal rain bands is debated in the literature, but beyond the present scope to further speculate here.

4 Synoptic Origins and Phenomenological Synthesis

The October 12th, 2023 event was detected as the strongest signal in an objective survey of three months of data (September–November 2023), Fig. 4a. This time series is averaged over a distinct geographical "hotspot" of NE-propagating wave events between 10°S and 30°S, 100°W to 80°W (not shown). While this region has relatively infrequent wave detections, compared to equatorial margins, it exhibits the most energetic and directionally organized wave packets in the basin. As recently emphasized by L. R. Allen et al. (2025), coherent, large-amplitude non-orographic gravity waves are episodic and rare, although fainter signatures are visually common. This spiky or episodic nature may suggest advective nonlinearity as the source, since momentum advection terms are quadratic in disturbance intensity.

The physical trigger of this extreme event appears to be an upper-tropospheric trough and jet streak in midlatitudes (Fig. 4b). Divergence at the 250 hPa level (blue-red colors) exhibits strong NW-SE elongated features in the region southwest of the low-level wave packet in Fig. 3, in and around a kinked trough at the entrance region of a subtropical jet streak. Such upper-level divergence is thought to be a marker of wave radiation by continuous adjustment to flow imbalance in highly curved, rapidly evolving synoptic flows. In their similar case study, G. Allen et al. (2012) displayed similarly banded residuals from a non-linear balance equation. Since those computations do not predict exact forcing patterns in terms of wavelengths, frequencies, and phase speeds, simple divergence suffices to illustrate the geometry of the banded wave emission.

As detailed by Zhang (2004), the flow in these jet-front systems undergoes a continuous "balanced adjustment" to restore equilibrium in the presence of advective balance disruption, emitting mesoscale gravity waves downward through the troposphere. Such an advective generation mechanism at upper levels has also been recently invoked for waves in dense surface pressure network observations of non-orographic storms (L. R. Allen et al., 2025). It is unclear whether vertical propagation is a more useful paradigm than the excitation of vertical "modes" (standing waves between the reflecting surface and the partly-reflecting tropopause) according to the projection of vertically deep forcings. Explanatory discussions of the scales and speeds of internal waves will be much better grounded when a lot more observations of this sort are available.

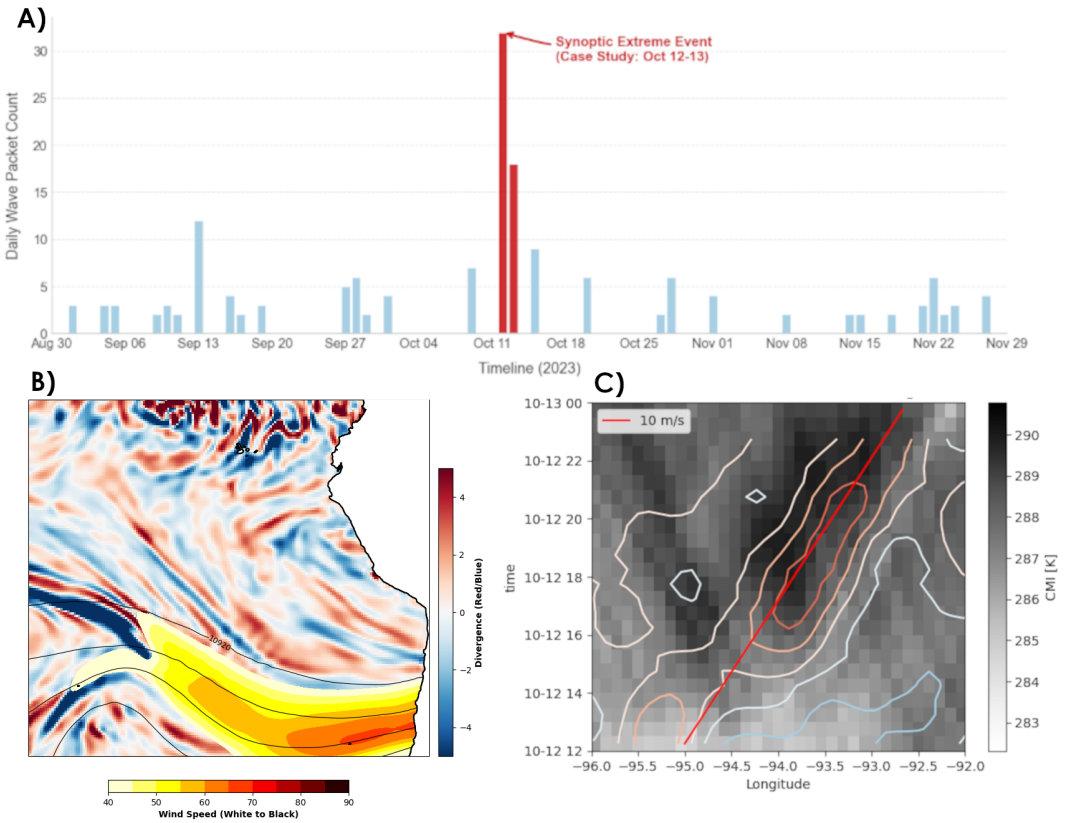


Figure 4. Phenomenological synthesis of the 12 October 2023 wave event. (a) Daily time series of automated wavelet detection count for northeastward propagating gravity waves in 10°S and 30°S, 100°W to 80°W. (b) 250 hPa divergence (blue-red), geopotential height (contours), and windspeed (yellow-orange), with South American landmass masked as white. (c) Time-longitude (Hovmöller) diagram of GOES-16 Band 13 brightness temperatures (grays) and divergence of low cloud-tracking winds (6 equally spaced contours from -0.2 to 0.2 $\times 10^{-4} s^{-1}$, red is positive), averaged over a 500 km wide SW-NE transect across the wave (location shown on Supporting Information videos S4 a-c).

Whatever the exact source mechanism, and whatever one's taste in vertical mode vs. vertically propagating ray descriptions, the wave is well observed in boundary layer cloudiness. A time-longitude (Hovmöller) diagram of T_b (gray shading in Fig. 4c) captures the north-eastward motion of a cloud-clearing wave (darker shading later in time). That clearing follows 1/2 wavelength behind a band of low-level wind divergence (red contours), deduced from a cloud-tracking wind analysis (method detailed briefly in Supporting Information S4). Besides this cloud-clearing seen directly in satellite data, a corresponding clearing is also seen in the ERA5 reanalysis model's low cloud field (Fig. S4

c,d), indicating that this wave's synoptic and mesoscale propagation as well as source characteristics are well handled by the model, despite its low cloud biases.

Wave speed is fairly clear on Fig. 4c: approximately 10 m/s (red line, corrected for slant orientation of the transect). This kinematic wind feature (divergence) is entirely independently measured, and corroborates very well the intrinsic phase speeds derived by our wavelet algorithm (Fig. 1c), for this strong case in a wind-perpendicular direction.

5 Summary and Perspectives

This 12 October 2023 wave event was found from a systematic, objective climatological screening procedure over the Southeast Pacific for September–November 2023. Three main wave sources are suggested by that 90-day survey (not shown): from the ITCZ in the deep tropics, from the western coast of South America, and from the subtropical - midlatitude weather of the South Pacific as in Fig. 4. For these several-km deep waves with their $o(10)$ m/s speeds, moisture flux convergence is a useful animation variable to see their lowest part, and gives a similar impression of these 3 main sources (not shown). From the latter source, a geographic hotspot between 15°S and 25°S presents the most energetic and directionally organized wave events in the basin. As in G. Allen et al. (2012), we suspect balance adjustment in highly curved troughs on the midlatitude jet to be the wave source.

Cloud brightness, our proxy for the marine inversion height, means we are fundamentally tracking *vertical displacements* (i.e., undulations in potential temperature θ surfaces), an indefinite integral of vertical velocity (w). Evaluating gravity waves via w, a common approach in diagnostics geared toward momentum fluxes like ($\overline{u'w'}$) which are the main importance of gravity waves for the middle atmosphere, relatively overemphasizes short, high-frequency waves by a factor of their frequency. Letting w determine contours or color scales risks dismissing broader features as negligible background, when displacement is the true "importance function" for wave-cloud interactions.

The vertical structure of these waves can be framed either as vertically propagating beams, or as discrete vertical modes (representing the interference of reflected beams). While both representations are valid, as explored numerically for a continuously stratified unbounded medium in (Mapes, 1998), the mode view better captures a fundamen-

tal feature of the profile of buoyancy frequency N for a moist adiabat (a good approximation for the tropical troposphere). N has a distinctive swoop to low values near 8 km altitude, where water vapor's saturation curve attenuates rapidly, below the tropopause where N increases sharply. This structure creates a mode or propagation cavity that is easily excited by a deep heating profile even without any special vertical wavelength to it. The peak susceptibility of this cavity is to waves with c_{int} of about 14 m/s for the deep tropics (Mapes, 2001, Fig. 8). The subtropics with their cooler temperatures have a similar but shorter N profile, and we hypothesize that momentum forcings such as balance adjustment, which also span deep layers of the troposphere, might similarly project onto this mode structure, perhaps explaining the prevalence of wave speeds in that speed range.

Besides their intrinsic dynamical interest, the displacement waves our method detects possess larger-scale importance to the cloud decks themselves. Even if kinematic perturbations are dynamically reversible, they can trigger irreversible microphysical transitions, such as cloud clearing or precipitation development and condensation nuclei washout, which can lead to the formation of pockets of open cells (POCs) (G. Allen et al., 2012), and thus to a net effect on the albedo of Sc cloud decks.

Certain limitations of this study must be acknowledged. The Stratocumulus-Topped Boundary Layer (STBL) is characterized by intense inherent mesoscale and cellular convection (Wood, 2012), including lines in bands of clouds, so not every comb shape implies a gravity wave. The c_{int} signal we measure is diluted by advecting brightness features with similar wavelength components (Supporting Text S1) .

Smoother, cleaner views of these waves may be available at a higher altitude: Poujol and Bony (2024) demonstrated the ability to measure vertical motions in clear air using water vapor absorption channels. Fusing these clear-air midlevel and cloudy-sky near-surface retrievals could enable a complete tomography-like reconstruction of the wave field across the depth of troposphere, measuring vertical wavelength directly as a powerful cross-check on the (linearly proportional) propagation speed. Might forward extrapolation allow predictions of internal-wave impacts such as coastal clearings or convective initiation?

By transforming archived satellite imagery into a dynamic wave catalog, the methodology described here could contribute to constraining the contribution of non-orographic

gravity waves to the global momentum budget in the middle atmosphere, a component that remains largely unparameterized in current climate models (Houtman et al., 2025). Others could use such a catalog to explore how wave-induced dynamics might perturb the fragile radiative equilibrium of marine boundary layers in a changing climate (Wood, 2012). Waves generated by deep convection and cyclones are seen in many cases (not shown here), so retrieving those could offer indirect inferences about vigorous imbalanced and transient dynamics hidden below high cloud shields. For all these reasons, the methods described here will be applied to larger datasets worldwide as the project progresses, with results reported as they emerge.

Open Research Section

The Matlab scripts used to generate results and figures in this paper are openly available in a public repository on Zenodo at `https://doi.org/10.5281/zenodo.20761882`. The GOES-16 Advanced Baseline Imager (ABI) Level 2 Cloud and Moisture Imagery Products (CMIP), specifically the Full Disk Band 13 brightness-temperature imagery used in this study, are publicly available from NOAA/NCEI at `https://doi.org/10.7289/V5736P36` and through the NOAA GOES public archive on AWS; these data were accessed on 27 August 2025. The NCEP/CPC L3 Half Hourly 4 km Global (60S–60N) Merged IR V1 product (GPM_MERGIR) is publicly available from NASA GES DISC at `https://doi.org/10.5067/P4HZB9N27EKU`; these data were accessed on 24 January 2026.

Acknowledgments

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